

Research Proposal

AI-Driven Transformation of Accounting Processes in Healthcare Organizations: Error Reduction, Process Control, and Cost Efficiency — Evidence from Cyprus and Greece

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Title and Research Field

The proposed dissertation, “AI-Driven Transformation of Accounting Processes in Healthcare Organizations: Error Reduction, Process Control, and Cost Efficiency — Evidence from Cyprus and Greece”, is located in the field of Accounting Information Systems (AIS) within Business Administration. It investigates how artificial intelligence transforms accounting and financial management processes in hospitals, focusing on three outcome dimensions: the reduction of human error in financial transactions, the improvement of process control and compliance, and measurable cost efficiency. The research will be supervised by Prof. Alexios Kythreotis, Associate Professor of Financial Accounting at the European University Cyprus.

Aims and Objectives

The dissertation pursues four interconnected objectives: (1) to identify and analyze the current state of accounting process automation and AI adoption in hospital accounting departments in Cyprus and Greece, with reference to two EUC-affiliated partner institutions — the American Medical Center in Nicosia and the European Interbalkan Medical Center in Thessaloniki; (2) to examine the relationship between AI adoption and the reduction of human error in financial transactions, including billing errors, payment processing mistakes, and compliance failures; (3) to assess how AI-based tools improve financial process control, audit trails, and regulatory compliance; and (4) to quantify the cost efficiency gains associated with AI adoption and derive evidence-based recommendations for healthcare managers, CFOs, and policymakers. The overarching research question asks: to what extent does the adoption of artificial intelligence in hospital accounting departments reduce human error, improve process control, and generate measurable cost efficiency gains — and how do these effects differ across the Cypriot and Greek healthcare contexts?

Importance of the Proposed Study

Hospital accounting departments are among the most error-prone administrative units in healthcare: manual data entry, fragmented information systems, and high transaction volumes produce billing errors, duplicate entries, and compliance failures with error rates in manual hospital billing estimated between 7% and 26%. These errors inflate operational costs, distort financial reporting, expose institutions to regulatory liability, and ultimately harm patients. Cyprus and Greece provide complementary and underexplored settings: the

Cypriot General Health System (GHS), introduced in 2019, has fundamentally restructured reimbursement and reporting requirements, while the Greek National Health System (ESY) presents a distinct, unevenly digitized institutional context. Despite the growing availability of AI-based accounting tools, no empirical study has systematically examined the accounting level of hospitals in the Eastern Mediterranean. The comparative cross-national design allows the analysis of how institutional context shapes the relationship between AI implementation and accounting process improvement — with direct policy relevance for both health systems. The study also has a translational dimension: its empirical mapping of error patterns and adoption barriers builds on a working prototype (HealthLedger AI) developed by the applicant, which automatically detects twelve types of billing errors in real hospital information system data.

Theoretical Background and Literature Review

The study draws on three complementary theoretical perspectives. Accounting Information Systems theory provides the lens for understanding how information technology transforms financial data collection, processing, and internal control effectiveness (Romney & Steinbart, 2021). Human error theory links accounting errors in complex, high-volume environments to manual data entry, multitasking, time pressure, and system fragmentation; recent evidence shows that AI adoption significantly reduces error frequency and improves fraud detection (Vrizaru, 2024). Transaction Cost Economics explains the conditions under which automation generates measurable efficiency gains (Tenhunen, 2025). The empirical literature documents rapid growth of AI in accounting and finance (Sanz Martín et al., 2025) and accelerating uptake of automation in healthcare administrative processes such as billing, claims management, and reconciliation (Nimkar et al., 2024): AI-enabled revenue cycle management reduces claim denials by 15–30%; most AI interventions in healthcare are cost-effective or cost-saving (Arab & Moosa, 2025); and intelligent automation reduces accounting errors by more than 50% (Cunha, 2022). Yet a structural gap persists: no study treats hospital financial management as a unified domain, a gap explicitly identified in the umbrella review by Clark et al. (2025), and research on small and medium-sized European healthcare markets is particularly scarce (Al-Hattami et al., 2025). This dissertation addresses both the thematic and the geographic gap.

Proposed Methodology

The dissertation adopts a quantitative research design combining a structured survey with secondary financial data analysis. In Phase 1, a questionnaire — built on validated scales from the AIS and technology adoption literature and piloted before full deployment — will be administered to accounting staff, finance managers, and CFOs at the two partner institutions. It measures current AI and automation adoption, perceived impact on error frequency, process control and cost efficiency, and organizational barriers and enablers.

Hypotheses will be tested using Structural Equation Modelling (SEM) with SmartPLS. In Phase 2, secondary financial data — billing and reimbursement error rates, error-correction costs, audit findings, and operational cost indicators — will be collected where access can be negotiated, providing objective validation of the survey findings and enabling comparison between institutions at different stages of AI implementation. All data collection will follow GDPR and the ethical guidelines of the European University Cyprus, with informed consent and formal data-sharing agreements. The timeline foresees coursework, literature review, instrument development and data access agreements in Year 1 (2027–2028); data collection, SEM analysis and a first journal article in Year 2 (2028–2029); and final analysis, a second article, and the dissertation defense in Year 3 (2029–2030). Expected contributions include the first systematic quantitative dataset on AI adoption in Cypriot and Greek hospital accounting, an integrated theoretical framework combining AIS theory with Transaction Cost Economics, and practical recommendations for hospital CFOs and policymakers in both health systems.